

# Luis Avalos

## Principal Software Engineer

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### SUMMARY

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Backend-focused developer with a strong emphasis on backend engineering while maintaining frontend proficiency. Hold a B.S. in Computer Science from the University of California, Riverside. Has engineered scalable cloud architectures and led multifaceted projects in remote/hybrid environments. Proven track record in designing cost-effective, high-performance systems that deliver significant cost savings (up to \$890K/year) and reduce latency and deployment times by 30%. Demonstrates expertise in constructing innovative distributed systems, defining technical strategy, and guiding agile teams toward high-quality outcomes.

### SKILLS

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- **Programming Languages:** Java, Python, C++, JavaScript, TypeScript
- **Frameworks & Libraries:** Spring Boot, Wildfly, Django, D3.js, React, FastAPI, Next, Angular, Redux
- **API Development:** RESTful APIs, GraphQL
- **Cloud & DevOps:** AWS(Cloudfront, EC2, Lambda, S3, API Gateway, RDS, CloudFormation), Google Cloud Platform(GCP), Azure, Docker, Kubernetes, Jenkins, GitLab CI, CI/CD Pipelines
- **Databases:** Amazon RDS, Apache Cassandra, PostgreSQL, NoSQL
- **Tools & Methodologies:** Git, Maven, Gradle, Agile, DevOps, Microservices Architectures, Data Modeling, Query Optimization, Algorithm Design & AI Training

### PROFESSIONAL EXPERIENCE

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#### DigitalOcean, Inc.

New York, NY

#### Principal Software Engineer - Distributed Systems

10/2022 - 12/2024

- **Project:** “Titan”
- **Objective:** Modernize a legacy monolithic backend into a scalable microservices architecture on AWS.
- **Responsibilities:**
  - Spearheaded architectural redesign and served as technical lead for a cross-functional team of 11 engineers.
  - Containerized microservices and coordinated their deployment via **Kubernetes**.
  - Established **CI/CD pipelines** (using **Jenkins/GitLab CI**) that reduced deployment cycles by 30%.
  - Partnered with product, operations, and design teams to ensure seamless integration with user-facing applications.
- **Impact:**
  - Reduced system latency by 30%, saved up to \$890K annually on data transfer and software delivery costs, and improved overall scalability.

#### Senior Backend Engineer

03/2019 - 10/2022

- Directed a small team to revamp distributed API gateways and integrate enhanced frontend monitoring solutions using **Java**, **Spring Boot**, and **AngularJS**.
- Reengineered caching and load-balancing strategies, which resulted in annual savings of up to \$500K and bolstered transaction throughput for systems managing over 1M daily transactions.
- Mentored team members on advanced microservices methodologies and best practices.

#### Backend Developer

07/2017 - 02/2019

- Constructed core microservices using **Java** and **Python** to support DigitalOcean’s developer platform.

- Integrated lightweight frontend components (leveraging **AngularJS**) into internal admin dashboards.
- Collaborated in a remote/hybrid team environment, building the foundation for scalable, high-availability systems.

## **Trissential**

**Rochester, MN**

### **Software developer(Java)**

**11/2023 - 06/2024**

- Upgraded a legacy healthcare system by migrating **Java 8** to **Java 17 (Spring Boot)**, improving maintainability, security, and performance, and ensuring a 100K+ lines of code transition without downtime.
- Refactored and redesigned backend APIs using **REST & GraphQL**, enabling modern frameworks to integrate seamlessly and improving system response time by 40%.
- Redesigned and developed user interfaces for healthcare applications using **React & Next.js**, improving UI performance and accessibility.
- Boosted frontend performance by implementing lazy loading, code splitting, and state management with **React Context & Redux**, improving page load speed by 35%.

## **Redhorse Corporation**

**Arlington, VA**

### **Artificial Intelligence/ML Engineer**

**07/2020 - 12/2020**

- Developed an AI-powered coding challenge evaluation system using **Python** and **JavaScript**, enhancing the accuracy of AI-generated solutions by 30%.
- Created an automated test framework with **FastAPI, Docker, and AWS**, generating diverse test cases to help AI handle edge cases more effectively, reducing incorrect responses by 40%.
- Optimized algorithm design and benchmarking methods to reduce code execution time by 25%.
- Improved AI chatbot explanations by providing structured feedback to ML engineers, making outputs more instructive and human-friendly.

## **Amazon Web Services(AWS)**

**Seattle, WA**

### **Backend Development Intern**

**06/2016 - 08/2016**

- Engineered Java microservices that increased processing efficiency by 25%.
- Built Python automation scripts to enhance infrastructure monitoring.
- Gained valuable experience in AWS cloud architecture and best practices in scalable system design.

## **EDUCATION**

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### **University of California, Riverside(UCR)**

**Riverside, CA**

### **Bachelor of Science in Computer Science**

**08/2013 - 05/2017**

- **GPA:** 3.7/4.0
- **Focus:** Algorithms, distributed systems, and cloud architectures.
- **Senior Capstone:** Designed a scalable RESTful API in Java and Python (Departmental Honors).
- Active participation in hackathons and coding clubs, developing an 80% backend and 20% frontend skill set.